

Exercise 1 (Matlab)

(a) Solve the following ODEs:

$$\ddot{x} + 4\dot{x} + 4x = e^{-t}, \quad x(0) = 1, \quad \dot{x}(0) = 1$$

$$\ddot{x} + 4x = t, \quad x(0) = 0, \quad \dot{x}(0) = 1$$

$$\ddot{x} + x = \sin t, \quad x(0) = 1, \quad \dot{x}(0) = 0$$

$$\ddot{x} + 4\dot{x} + 3x = 2e^{-3t}, \quad x(0) = 0, \quad \dot{x}(0) = -1$$

Solution 1:

$$t \exp(-t) + t \exp(-2t) - \exp(-t) (t - 1)$$

Solution 2:

$$t \sin(2t) - \frac{3}{4} + \frac{t^2}{8}$$

Solution 3:

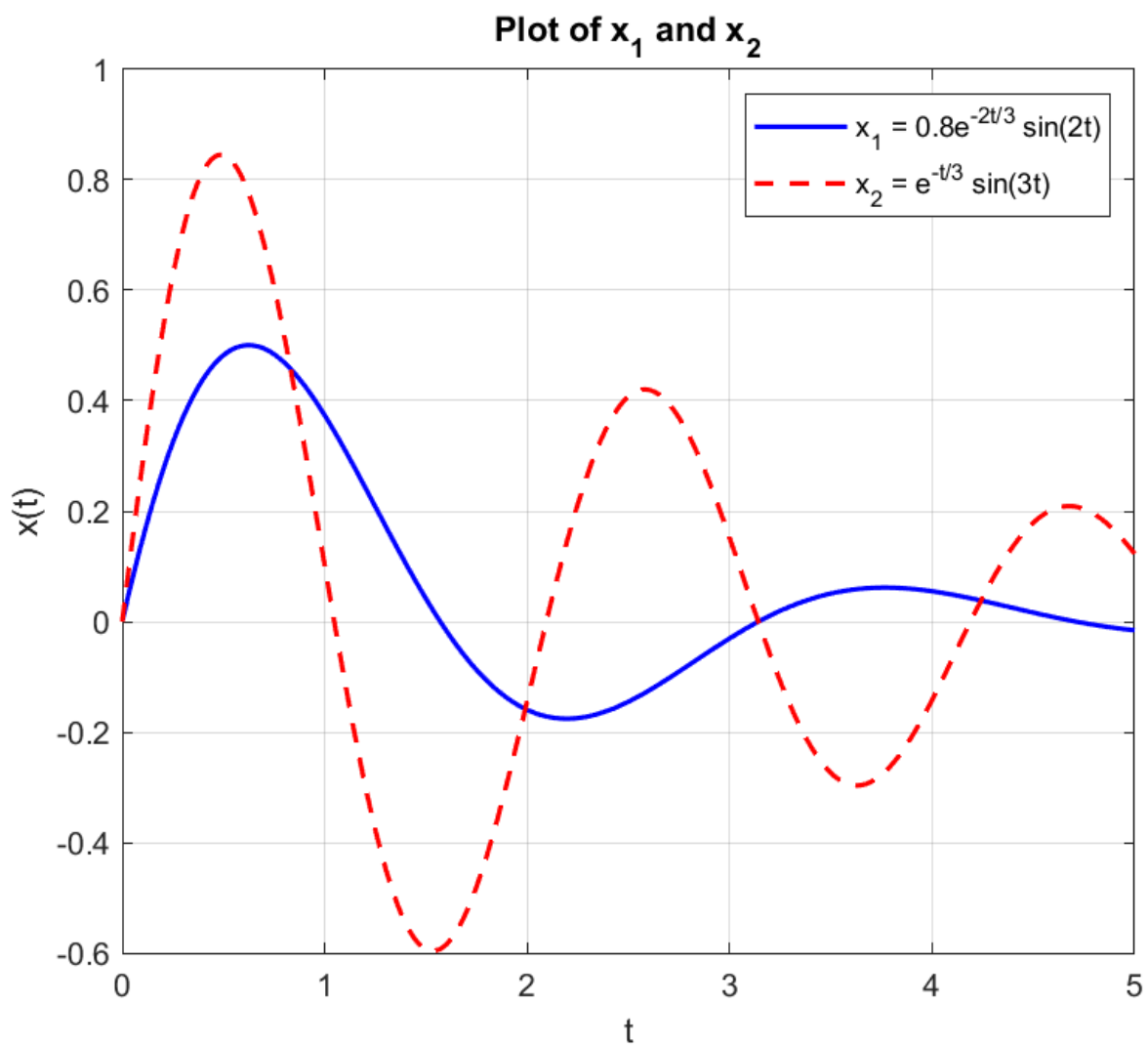
$$\cos(t) - \frac{\sin(3t)}{8} + \frac{3 \sin(t)}{8} - \cos(t) \sqrt{\frac{t}{2}} - \frac{\sin(2t)}{4} \sqrt{t}$$

Solution 4:

$$-t \exp(-3t)$$

(b) Plot the functions

$$(x_1 = 0.8e^{-2t/3} \sin 2t) \text{ and } (x_2 = e^{-t/3} \sin 3t \quad 0 \leq t \leq 5)$$



(c) Plot the function

$(y(x, t) = e^{-2x} \cos(t + x))$ vs $(0 \leq x \leq 2)$ for $(t = 0, 1, 2, 3)$

Plot of $y(x,t) = e^{-2x} \cos(t+x)$ for various t

